

SENTINEL ONE

One Guardian. Every Digital Threat.

PROJECT SUMMARY

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Hackathon Project | Cybersecurity & AI

Executive Summary

SENTINEL ONE, developed by team Cyber Sentinels, is a Hybrid AI Detection Engine built to protect users from digital fraud before they interact with suspicious content. Rather than reacting after a scam has already caused harm, the platform proactively detects fraud emails, scam SMS, fake banking websites, malicious QR codes, and unknown scam calls, verifying threats and alerting users before they click, scan, answer, or pay. The system combines multiple AI technologies, including Natural Language Processing, Computer Vision, Optical Character Recognition, and Speech Keyword Detection, into a single Hybrid AI Detection Engine, and pairs this with Explainable AI so that every alert is easy to understand. A strong emphasis on privacy ensures that only calls from unknown numbers are analyzed, saved contacts are never monitored, and all analysis takes place on-device without storing or uploading conversations. Designed for banks, FinTech companies, UPI users, digital banking customers, senior citizens, and government financial services, SENTINEL ONE brings together detection, verification, and protection across every major digital banking channel into one unified, intelligent, and privacy-first security solution.

1. Project Overview

SENTINEL ONE is a Hybrid AI Detection Engine designed to protect users from digital fraud before they interact with suspicious content. Its main purpose is to shift digital security from a reactive model, where action is taken only after fraud has occurred, to a proactive model that identifies and neutralizes threats in advance. By combining multiple detection technologies into a single platform, SENTINEL ONE aims to safeguard users across the everyday digital channels most commonly exploited by scammers, including emails, SMS messages, websites, QR codes, and phone calls.

2. Problem Statement

Digital fraud today takes many forms, and users are exposed to threats through several different channels at once. SENTINEL ONE addresses the following key threats:

- Fraud emails designed to trick users into revealing sensitive information
- Scam SMS messages containing malicious links or false urgency
- Fake banking websites created to steal login credentials
- Malicious QR codes that redirect users to fraudulent payment pages
- Unknown scam calls attempting identity theft or financial fraud

Traditional security approaches are largely reactive, responding to fraud only after it has already taken place. This leaves a critical gap, since by the time a threat is identified, the user may have already clicked a malicious link, scanned a harmful QR code, answered a scam call, or made a fraudulent payment. A proactive approach is essential to intercept threats before users take an action that could compromise their finances or personal data.

3. Proposed Solution

SENTINEL ONE is proposed as a unified intelligent security platform that brings together detection capabilities for multiple types of digital threats under a single system. Its core approach follows a simple and consistent cycle:

Detect → Verify → Alert → Protect

This approach ensures that potential threats are identified early, verified for authenticity, and communicated to the user through a clear alert, so that protective action can be taken before the user clicks a suspicious link, scans an untrusted QR code, answers a scam call, or completes a payment.

4. Key Features

- Fraud email detection
- Scam SMS detection
- Fake banking website detection
- Malicious QR code detection
- Unknown scam call detection
- Identity verification
- Real-time threat assessment
- Explainable security alerts
- AI recommendations
- Risk-level indication

5. Privacy-First Innovation

SENTINEL ONE is built around the principle that user privacy must never be compromised in the process of delivering security. The platform follows these core privacy principles:

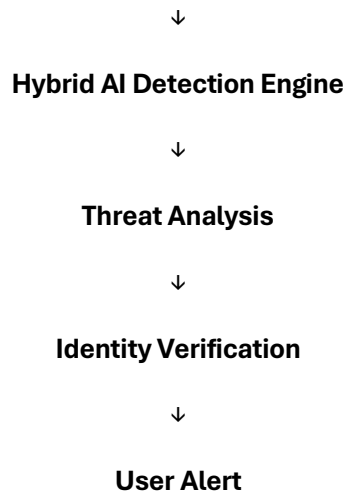
- Only incoming calls from unknown or unsaved numbers are analyzed
- Saved contacts are never monitored
- Analysis happens completely on-device
- Conversations are never stored or uploaded

A privacy-first design is essential for any digital security solution because users are far more likely to trust and adopt a system that protects them without invading their personal space. By limiting analysis strictly to unknown numbers and processing data on-device, SENTINEL ONE ensures that everyday communications with trusted contacts remain completely private, while still delivering strong protection against genuine threats.

6. System Workflow

The overall workflow of SENTINEL ONE follows a clear, sequential process:

Email / SMS / Unknown Call / QR Code / Website



Each incoming item, whether an email, SMS, unknown call, QR code, or website, first enters the Hybrid AI Detection Engine. The engine performs a threat analysis to assess whether the source shows signs of fraud. This is followed by an identity verification step to confirm the legitimacy of the sender or source. Finally, the system generates a user alert, informing the user of the outcome so they can decide how to proceed safely.

7. Hybrid AI Detection Engine

A key strength of SENTINEL ONE is that it combines multiple AI technologies rather than relying on a single AI model. This hybrid approach allows the system to handle different types of threats using the most suitable technology for each channel. The technologies used include:

- Artificial Intelligence
- Machine Learning
- Natural Language Processing (NLP)
- Computer Vision
- Optical Character Recognition (OCR)
- Speech Keyword Detection
- URL Reputation Analysis
- Phishing Detection Engine
- Cloud Threat Intelligence
- Explainable AI

Technology Mapping

- NLP → Email & SMS Analysis
- Computer Vision + OCR → QR Verification
- URL Analysis → Website Verification
- Speech Analysis → Unknown Call Detection

8. Detection Results

SENTINEL ONE classifies every analyzed item into one of three security outcomes, each accompanied by a clear status, reason, and recommended action:

Outcome	Status	Reason	Recommended Action
SAFE	Verified Source	Trusted sender or website	Safe to continue
SUSPICIOUS	Potential Risk	Unverified source or unusual activity	Verify before proceeding
FRAUD DETECTED	High Risk	Known phishing or scam indicators	Block immediately and report

9. Competitive Advantage

SENTINEL ONE offers a clear advantage over traditional security approaches by addressing multiple digital threat channels through a single, intelligent platform:

Traditional Security	SENTINEL ONE
Password Protection	Hybrid AI Detection
OTP Verification	Privacy-First Design
Manual Verification	Multi-Channel Detection
Reactive Security	Proactive Prevention
No QR Protection	QR Code Detection
Limited Website Detection	Website Protection
Limited Email Security	Email Protection
Limited SMS Security	SMS Protection
No Scam Call Detection	Scam Call Detection
Limited AI Explanation	Explainable AI

Competitive Edge: *One intelligent platform protecting every digital banking channel.*

10. Prototype

The prototype for SENTINEL ONE demonstrates the end-to-end workflow of the system through the following screens:

- Dashboard
- Unknown Call Detection

- Email Analysis
- SMS Analysis
- QR Verification
- Website Verification
- Final Security Alert Dashboard

Dashboard Features

- Real-time threat score
- AI recommendations
- Risk level indicator
- Explainable alerts
- Security history

11. Target Users

SENTINEL ONE is designed to serve a wide range of users across the digital banking ecosystem:

- Banks
- FinTech Companies
- UPI Users
- Digital Banking Customers
- Senior Citizens
- Government Financial Services

Each of these groups faces regular exposure to digital fraud attempts, from phishing emails targeted at bank customers to scam calls aimed at senior citizens. A unified, proactive detection platform such as SENTINEL ONE is directly relevant to protecting all of these users across the channels they rely on most.

12. Future Scope

The following areas have been identified as future development opportunities for SENTINEL ONE:

- Regional Language Scam Detection
- Voice Biometrics
- Bank API Integration
- Dark Web Identity Monitoring
- Cross-Platform Protection
- Real-Time Fraud Intelligence

13. Conclusion

SMS, QR codes, websites, and unknown phone calls, before fraud has the chance to happen. By combining a Hybrid AI Detection Engine with Explainable AI and a proactive approach to security, the platform builds user trust and works toward making digital banking safer for everyone.

“SENTINEL ONE – One Guardian. Every Digital Threat.”